AQ11.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y) = x\log(y)$f(x, y) = x log(y) and $Hf(x, y)$Hf(x, y) denote the Hessian matrix of ff at (x, y)(x, y). Which of the following options are correct?

- [ ] Rank of $Hf(0, 1)$Hf(0, 1) is 2.
- [ ] Rank of $Hf(0, 1)$Hf(0, 1) is 1.
- [ ] $(0, 1)$(0, 1) is a local minimum.
- [ ] $(0, 1)$(0, 1) is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Rank of $Hf(0, 1)$Hf(0, 1) is 2.
$(0, 1)$(0, 1) is a saddle point.

1 point

Choose the correct statements about $f(x, y) = x^2 + axy + by^2$f(x, y) = x2 + axy + by2.

- [ ] $(0, 0)$(0, 0) is a saddle point of ff iff $4b - a^2 < 0$4b − a2 < 0.
- [ ] $(0, 0)$(0, 0) is a saddle point of ff iff $4b - a^2 > 0$4b − a2 > 0.
- [ ] There are infinitely many possible pairs (a, b)(a, b) for which ff has local minimum at $(0, 0)$(0, 0).
- [ ] If $a = b = 1$a = b = 1 then ff has a local maximum at $(0, 0)$(0, 0).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(0, 0)$(0, 0) is a saddle point of ff iff $4b - a^2 < 0$4b − a2 < 0.
There are infinitely many possible pairs (a, b)(a, b) for which ff has local minimum at $(0, 0)$(0, 0).

1 point

Consider the following function

$$f(x, y) = x^2 + xy + y^2 + 2x - 2y + 5$$f(x, y) = x2 + xy + y2 + 2x − 2y + 5

Which of the following option(s) is (are) true?

- [ ] $(-2, 2)$(−2, 2) is a critical point of ff.
- [ ] $(2, -2)$(2, −2) is a critical point of ff.
- [ ]

Determinant of the Hessian matrix at point (x, y)(x, y) is 33.

☐

ff has a local minimum at point $(-2, 2)$(−2, 2).

☐

$(2, -2)$(2, −2) is a saddle point of ff.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-2, 2)$(−2, 2) is a critical point of ff.

Determinant of the Hessian matrix at point (x, y)(x, y) is 33.

ff has a local minimum at point $(-2, 2)$(−2, 2).

For what value of kk is the Hessian test inconclusive for the function $f(x, y) = x^3 + kxy + y^3$ f(x, y) = x3 + kxy + y3 at (0, 0)(0, 0)?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2

1 point

Choose the correct statements about $f(x, y) = 2x^3 - 3x^2 y - 12x^2 - 3y^2$ f(x, y) = 2x3 − 3x2y − 12x2 − 3y2.

☐

ff has 33 critical points.

☐

(0, 0)(0, 0) is a local maximum.

☐

(0, 0)(0, 0) is a saddle point.

☐

$(-4, -8)$(−4, −8) is a saddle point.

☐

$(-4, -8)$(−4, −8) is a local minimum.

☐

ff has a global minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

ff has 33 critical points.

(0, 0)(0, 0) is a local maximum.

$(-4, -8)$(−4, −8) is a saddle point.

1 point

Choose the correct statements about $f(x, y) = x^8 - 2x^4 y + y^3 - y$f(x, y) = x8 − 2x4y + y3 − y.

☐

The Hessian matrix of ff at point (x, y)(x, y) is $\begin{bmatrix} 56x^6 - 24x^2 y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$[56x6 − 24x2y −8x3 −8x36y].

☐

The Hessian matrix of ff at point (x, y)(x, y) is $\begin{bmatrix} 56x^6 - 24x^2 y & 8x^3 \\ 8x^3 & 6y \end{bmatrix}$[56x6 − 24x2y8x38x36y].

☐
ff has four critical points.

☐
$(1, 1)$(1, 1) is a point of local minimum of ff.

☐
$(1, 1)$(1, 1) is a point of local maximum of ff.

☐
Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$(0, 3

√
1).

No, the answer is incorrect.
Score: 0
Accepted Answers:

The Hessian matrix of ff at point (x, y)(x, y) is $\begin{bmatrix} 56x^6 - 24x^2 y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$[56x6 − 24x2y−8x3−8x36y].

ff has four critical points.
$(1, 1)$(1, 1) is a point of local minimum of ff.
Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$(0, 3

√
1).

If MM denotes the local maximum value of the function $F(a, b) = \int_a^b 3 - 2x - x^2 \, dx$F(a, b) = ∫ab

3 − 2x − x2 dx over all possible values of aa and bb then find the value of $3M$3M. Note that the function has only one local maximum and you have to find that.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 32
1 point
1 point
Consider a function $f(x, y)$f(x, y). Let v_1, v_2, v_3v1, v2, v3 be the critical points of the function ff and H_k Hk be the Hessian matrix of the function at the point v_k vk.Then which of the following options is(are) true?

☐
If $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$H1 = [1−302], then v_1 v1 is a saddle point.

☐
If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$H2 = [1312], then v_2 v2 is a saddle point.

☐
If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$H3 = [3−311], then v_3 v3 is a point of local extremum.

☐
If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$H1 = [2002], then v_1 v1 is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$H2 = [1312], then v_2 v2 is a saddle point.

If H_3 = [3 1 −3 1]H3 = [3−311], then v_3 v3 is a point of local extremum.

If H_1 = [2 0 0 2]H1 = [2002], then v_1 v1 is a point of local minimum.

Check Answers

Your score is: 0/8